Exact weighted monomial counts

Prime norm study, 14 November 2026

Source freeze `740732e95389e5185ed9a6547c18d8d889a6c580`

Result. For positive natural weights $w_1, \dots, w_v$, let $M(w, D)$ count *every* exponent vector $e \in \mathbb{N}^v$ with $\sum_i w_i e_i \le D$. The new Lean module establishes

$$M(w,D)\prod_i w_i \;\le\; \binom{D+\sum_i w_i}{v}, \qquad M(w,D)\prod_i w_i\, v! \;\le\; (D+\sum_i w_i)^v.$$

For consecutive weights this gives

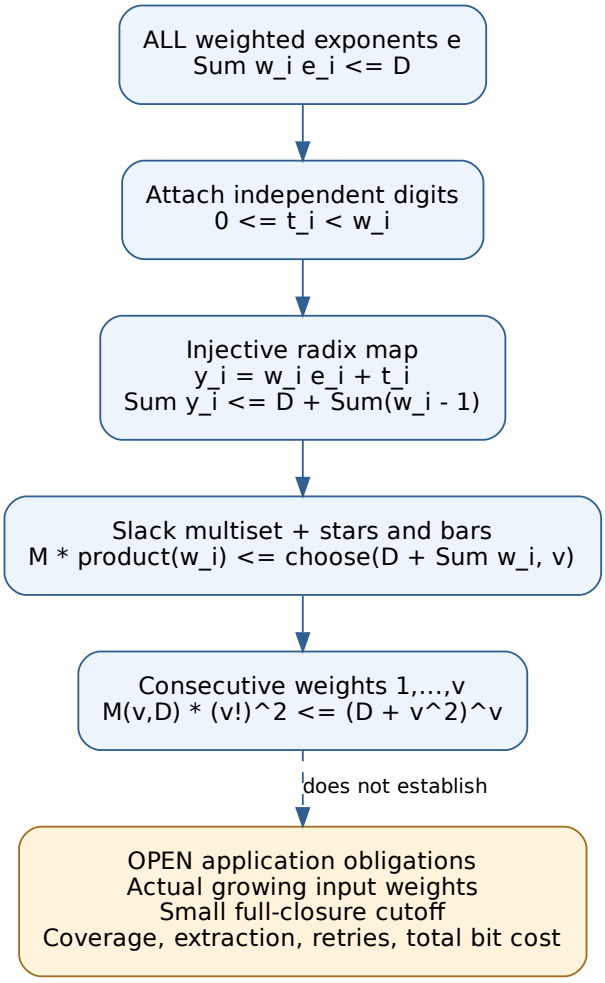
$$\boxed{M(v,D)(v!)^2 \le (D+v^2)^v.}$$

PCC-20261114-cc1d70 / 621b19 / 9932d5; unsigned [pcc:unverified]. The exact binomial inequality is retained, rather than only the coarser square bound. Zero variables and zero budget are included.

Proof mechanism. Attach digits $0 \le t_i < w_i$ and set $y_i = w_i e_i + t_i$. Euclidean division recovers both e_i and t_i, so the map is injective. Also $\sum_i y_i \le L := D + \sum_i (w_i - 1)$. Encode y as a multiset with y_i copies of label i and $L - \sum_i y_i$ copies of an extra slack label. Its cardinality is L; counts of the ordinary labels recover y. Mathlib stars-and-bars counts these multisets by $\binom{L+v}{v}$. Positive weights give $L + v = D + \sum_i w_i$. The descending-factorial bound supplies the second inequality. For weights $1, \dots, v$, the product is $v!$ and $\sum_i w_i \le v^2$.

Completeness of the counted set. The executable finite set is a box $0 \le e_i \le D$ filtered by weighted degree. The theorem `mem_monomials` proves equivalence to the weighted inequality for positive weights. No support list or moment oracle is supplied.

Application boundary. This is finite combinatorics, with no claim of novelty for the elementary argument. It does not prove that the growing norm inputs have this grading or that their full iterated closure stabilizes at a small cutoff. Public base coverage, signed output, extraction, failed attempts and complete expected bit cost remain explicit application obligations. An original $W_{\text{new}} = W_{\text{base}} + CH$ jet map is not established by the counting theorem.



Evidence and sources. Source: `lean/NormWeightedCount.lean`, ten authored theorems; exact declaration hashes in `weighted-count-evidence/claim-pins.json`. Build drafts 001 and 002 failed on elaboration/API usage and remain archived with their source; build003 exits0 without warnings. Gate and fresh replay receipts are recorded in the companion README. Unsigned claims do not assert signed standing.

Pinned Mathlib sources:
`Data/Sym/Card.lean`
`Data/Nat/Choose/Bounds.lean`
`Data/Nat/Factorial/BigOperators.lean`.

Prior-work review read actual P1514 producer/audit and P1536 norm-jet audit at crypto-autoresearcher `14f7a739d`. P1514's dense cutoff was sufficient, not a universal minimum. P1536's exact norm-jet inverse lacked a cheap constructor. Neither audit is an experiment measuring early weighted closure. KB tools and wrapper executable were unavailable.

No native fixture, solver, timing campaign, paid job or Python cryptomath was run. No canonical performance graph changes: no measured performance result is added.